

EFFECT OF
LIGNOCAINE &
PROCAINE ON
FROG'S LEGS

OBJECT

- **TO STUDY THE EFFECTS OF LIGNOCAINE AND PROCAINE ON THE LEGS OF A FROG.**

REQUIREMENTS

- Pithing Needle
- Syringe (2cc),
- frog,
- beakers,
- hot water (60 ° C)
- Thermometer
- drugs (Lignocaine, Procaine).

THEORY

- **LOCAL ANESTHETICS:** are drugs that block nerve conduction when applied locally to nerve tissue in an appropriate concentration. A local anesthetic in contact with a nerve trunk can cause both sensory and motor paralysis in the area innervated. The action of local anesthetics is completely reversible.

CLASSIFICATION OF LOCAL ANESTHETICS:

- **SURFACE ANESTHETICS:** are the drugs which, when applied on skin or mucous membrane, produce loss of sensation by depressing sensory nerve endings e.g.; **Lignocaine, Benzocaine** and **Orthocaine**. Lignocaine is also used in infiltration and spinal anesthesia. Benzocaine and Orthocaine are insoluble powders that may be blown into open wounds and painful throat.

-
- **INFILTRATION ANESTHETICS:** are drugs which produce anesthesia when injected into tissues in a suitable concentration. e.g.; **Procaine, Lignocaine**. About 1% Procaine acts directly on sensory organs or on the smallest branch of sensory nerves.
 - **CONDUCTION ANESTHETICS:** are drugs which block the nerves and nerve plexuses when injected into the neighborhood of a nerve supplying an area e.g.; **Amethocaine**, and **Procaine** (concentration of about 2% Procaine is required).
 - **REGIONAL ANESTHETICS:** are combinations of infiltration and conduction anesthetics e.g.; **Procaine**

-
- **SPINAL ANESTHETICS:** are drugs which are injected into the lumbar sub-arachnoid space below the termination of the spinal cord (i.e the L 3-4 inter-vertebral space) to block the conduction of sensory nerve impulses e.g.; **Tetracaine** and **Cinchocaine**. Tetracaine is also suitable for injections into mucous membranes while Cinchocaine is absorbed from mucous membrane and produces more prolonged effects.
 - **Disadvantages of Spinal Anesthetics:**
 - Sometimes produce inflammation and infection at the site of injection.
 - Liable to produce precipitation from watery solution unless kept in an acidic medium.

PROCEDURE

- Pith the given frog to destroy its medulla oblongata.
- Take a normal, baseline response by dipping both legs in hot water at 60°C.
- Put the right leg in a beaker containing Lignocaine solution for 5 minutes & then note the response in the right leg by dipping it in hot water.
- Inject 0.25cc Procaine in left leg in the vicinity of sciatic nerve, wait for few minutes and again see the response in hot water.

Procedure	Response in	Response in	Conclusion
	Right leg	Left leg	
Dip both legs in hot water.	+	+	-----
After dipping right leg in Lignocaine solution	-	+	Lignocaine is surface anesthetic.
After injecting 0.25 cc Procaine in left leg	+	-	Procaine is infiltration anesthetic.

RESULT

- Lignocaine is surface anesthetic while procaine is infiltration anesthetic.